Optical Observation and Inverse Problems for a Vibrating Membrane

An expository note with proofs and explicit examples

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Abstract

A reflected laser spot provides an indirect observation of membrane motion. We derive a simple optical linearization, identify its measurement operator, and distinguish initial-state reconstruction from source reconstruction. We prove an eigenspace criterion and an explicit Gramian formula for finite-mode reconstruction. For a known spatial source profile, we prove a Volterra reconstruction theorem with an optimal two-derivative loss in its stated Sobolev scale. For a uniformly driven circular membrane observed through a circular averaging aperture, we compute the exact Laplace transfer function and prove the precise observation time needed for uniqueness, using the Titchmarsh convolution theorem. A smooth averaging weight supplies a complementary example with no bounded inverse in any finite-order Sobolev data norm. All reconstruction statements refer to the specified linear observation models. The note makes no claim of originality.

Reading prerequisites. Sobolev spaces, the spectral theorem for a positive self-adjoint operator with compact resolvent, and elementary ordinary differential equations. The Titchmarsh convolution theorem is stated and cited when used. The special-function identities needed for the disk calculation are derived below, except for the standard large-argument Bessel asymptotic.

Main conclusions. Finite-time optical data may fail to distinguish exact modes; an injective data map may nevertheless have an unbounded inverse. The source class and the sensor must therefore be specified before posing a reconstruction problem. Uniform pressure is a useful test case: it defeats the simplest instantaneous nondegeneracy condition, yet an off-centre finite aperture still yields uniqueness after an exactly calculable delay.

Contents

1 From reflection geometry to an observation operator

Let a reference membrane occupy a bounded domain $\Omega \subset \mathbb{R}^2$. Its small transverse displacement is a scalar function $u(x,t)$, and its graph is

$$X(x,t) = (x, u(x,t)).$$

This graph description assumes that slopes remain finite and that the optical ray intersects the intended part of the surface. Its upward unit normal is

$$n(x,t) = \frac{(-\nabla u(x,t), 1)}{\sqrt{1+|\nabla u(x,t)|^2}}.$$

For a unit incoming direction d_{in}, specular reflection gives

$$d_{\text{out}} = d_{\text{in}} - 2(d_{\text{in}} \cdot n)n. \tag{1.1}$$

Proposition 1.1 (A calibrated optical linearization)**.** *Assume normal incidence* $d_{\text{in}} = (0,0,-1)$ *along the vertical line through* x_**, and a screen in the plane* $z = L$*. Put* $g(t) = \nabla u(x_*, t)$ *and* $h(t) = u(x_*, t)$*. If* $L > h(t)$ *and* $|g(t)| < 1$*, the horizontal screen position is*

$$\gamma(t) - x_* = -\frac{2(L - h(t))}{1 - |g(t)|^2} g(t). \tag{1.2}$$

The differential of this map at the flat equilibrium is

$$D\gamma(0)[u] = -2L\nabla u(x_*, t). \tag{1.3}$$

Proof. Equation (1.1) gives

$$d_{\text{out}} = \frac{(-2g, 1-|g|^2)}{1+|g|^2}.$$

The reflected ray starts at (x_*, h). Dividing its horizontal direction by its vertical direction and multiplying by $L - h$ proves (1.2). Differentiating at $(h,g) = (0,0)$ proves (1.3). For $|g| \le \eta < 1$, the difference from the linear term is bounded by a constant times $|h||g| + L|g|^3$. □

After subtracting the equilibrium spot and dividing by the known factor $-2L$, the idealized data are $y(t) = \nabla u(x_*, t)$. A finite-area effective observation has the form

$$Cu = \int_\Omega w(x)\nabla u(x)\,dx \in \mathbb{R}^2, \qquad w \in L^2(\Omega). \tag{1.4}$$

A single directional channel is $C_e u = \int w\, e \cdot \nabla u$ for a fixed unit vector e.

Modeling boundary. Formula (1.4) is an assumed, calibrated area-averaging model. A rigid mirror glued to a membrane can have additional translational and rotational dynamics, and its tilt is not automatically the average gradient of the bare membrane. That coupling must be modeled if it is appreciable. Likewise, illuminating an open set and recording only its centroid does not provide the spatially resolved field $\nabla u|_W$.

Remark 1.2 (Time information and regularity)**.** A parameterized curve $\gamma(t)$ retains timing, frequency and phase. Its derivative is determined by exact knowledge of the curve and cannot remove an existing kernel of the data map. A point gradient is not defined for general H^1 functions in two dimensions. Point observations below are used only for smooth or finite-mode solutions. In contrast, (1.4) is bounded on $H_0^1(\Omega)$:

$$|Cu| \le \|w\|_{L^2} \|\nabla u\|_{L^2}.$$

2 The wave equation and the unknowns

For the main results, Ω is bounded with smooth boundary, $c > 0$ is constant, and

$$H = L^2(\Omega), \quad A = -c^2\Delta_D, \quad V = D(A^{1/2}) = H_0^1(\Omega).$$

The Dirichlet operator is positive self-adjoint with compact resolvent. Equip V with $\|u\|_V = \|A^{1/2}u\|_H = c\|\nabla u\|_2$, and set

$$\mathcal{H} = V \times H, \qquad \|(u,v)\|_{\mathcal{H}}^2 = \|A^{1/2}u\|_H^2 + \|v\|_H^2.$$

We use the bounded operators

$$P(t) = \cos(tA^{1/2}), \qquad S(t) = A^{-1/2}\sin(tA^{1/2}).$$

Proposition 2.1 (Energy evolution)**.** *For $(u_0, u_1) \in \mathcal{H}$ and $f \in L^1(0,T;H)$, the energy solution of $u_{tt} + Au = f$ is*

$$u(t) = P(t)u_0 + S(t)u_1 + \int_0^t S(t-s)f(s)\,ds. \tag{2.1}$$

It belongs to $C([0,T];V) \cap C^1([0,T];H)$, is unique, and satisfies

$$\|(u(t), u_t(t))\|_{\mathcal{H}} \le \|(u_0,u_1)\|_{\mathcal{H}} + \int_0^t \|f(s)\|_H\,ds.$$

For $f = 0$, the energy norm is conserved.

Proof. In an orthonormal eigenbasis $A\phi_j = \lambda_j\phi_j$, each coefficient solves $q_j'' + \lambda_j q_j = f_j$. The scalar variation-of-constants formula yields (2.1). In the coordinates $(\sqrt{\lambda_j}q_j, q_j')$, the homogeneous evolution is an orthogonal rotation. Summing its squared norm over j proves conservation and defines a strongly continuous unitary group $U(t)$ on $\mathcal{H}$. The forcing enters as $(0, f)$; applying Minkowski's inequality to the variation-of-constants formula on $\mathcal{H}$ proves the estimate, convergence and continuity. Applying the same formula to a difference proves uniqueness. $\square$

Fix $C \in \mathcal{L}(V, \mathbb{R}^m)$. There are two separate inverse problems:

- *Initial-state reconstruction:* f is known and (u_0, u_1) is unknown. Subtract the known solution with zero initial state, reducing to $f = 0$.
- *Source reconstruction:* the initial state is known and $f(x,t) = b(x)a(t)$, where b is known and the scalar function a is unknown. Subtract the known homogeneous evolution, reducing to zero initial state.

Recovering an arbitrary unknown $f(x,t)$ is a different and much less determined problem. Known membrane parameters and boundary conditions do not specify either the initial state or the forcing. The general operator framework is treated in [1]; separated-source problems also occur in [2].

3 Exact obstructions to reconstruction

Example 3.1 (Invisible disk modes)**.** On $B_R(0)$, radial Dirichlet eigenfunctions are

$$\phi_{0,k}(x) = J_0(j_{0,k}|x|/R).$$

Their gradient at the centre vanishes. Thus $u(x,t) = \phi_{0,k}(x)\cos(cj_{0,k}t/R)$ is invisible to central point-slope observation. Under (1.2) its screen spot is stationary as well.

More strongly, for each integer $m \geq 2$, the smooth disk eigenfunction

$$\phi_{m,k}(r,\theta) = J_m(j_{m,k}r/R)\cos(m\theta)$$

satisfies $\phi_{m,k}(0) = 0$ and $\nabla\phi_{m,k}(0) = 0$. Indeed, $J_m(z) = z^m/(2^m m!) + O(z^{m+2})$, so its leading term is a homogeneous polynomial of degree m. These modes are invisible to any central observation depending only on the first jet $(u, \nabla u)$. These are failures of uniqueness, independent of observation noise.

Proposition 3.2 (An unrestricted source is not identifiable). *Let $W \subset \Omega$ and suppose $\Omega \setminus \overline{W}$ contains a nonempty open subset. Even with zero initial data, knowledge of the complete displacement on $W \times (0,T)$ cannot determine an arbitrary space-time source.*

Proof. Choose a nonzero $v \in C_c^\infty((\Omega \setminus \overline{W}) \times (0,T))$ and let $f = v_{tt} + Av$. Then $u = v$ satisfies the wave equation, zero initial and boundary data, and $u|_{W\times(0,T)} = 0$. The source is nonzero, since otherwise Proposition 2.1 would imply $v = 0$. The zero source and this source therefore produce identical observations. □

Proposition 3.3 (Compactness of bounded finite-channel observation). *Let $U(t)$ be the homogeneous wave group on the infinite-dimensional space $\mathcal{H}$, and put $\mathcal{C}(u,v) = Cu$. Then*

$$\mathcal{O}_T : \mathcal{H} \longrightarrow L^2(0,T;\mathbb{R}^m), \qquad \mathcal{O}_T z = \mathcal{C}U(\cdot)z$$

is compact. Consequently there is no constant K such that

$$\|z\|_{\mathcal{H}} \leq K\|\mathcal{O}_T z\|_{L^2(0,T)} \quad \text{for every } z \in \mathcal{H}. \tag{3.1}$$

Proof. Represent the components of $\mathcal{C}$ by vectors $h_1, \ldots, h_m \in \mathcal{H}$. The components of $\mathcal{C}U(t)z$ are $\langle z, U(t)^* h_j\rangle$. Each map $t \mapsto U(t)^* h_j$ is norm-continuous, uniformly on $[0,T]$. Approximate these maps uniformly by step functions. The resulting operators from $\mathcal{H}$ to $L^2(0,T;\mathbb{R}^m)$ have finite rank and converge to $\mathcal{O}_T$ in operator norm. Thus $\mathcal{O}_T$ is compact. If (3.1) held, a bounded inverse on its range would turn this compact map into the identity on $\mathcal{H}$, contradicting infinite dimensionality. □

Compactness does not imply noninjectivity. Proposition 3.3 rules out uniform energy-norm stability for these sensors. It does not settle uniqueness, and it does not apply directly to an unbounded point or boundary observation operator.

4 Finite-mode initial-state reconstruction

Choose a finite set $\mathcal{L}$ of positive eigenvalues and put

$$V_N = \bigoplus_{\lambda \in \mathcal{L}} E_\lambda, \qquad E_\lambda = \ker(A - \lambda I).$$

Assume $(u_0, u_1) \in V_N \times V_N$ and $f = 0$.

Theorem 4.1 (Eigenspace criterion). *For every $T > 0$, the initial state is uniquely determined by $y(t) = Cu(t)$ on $(0,T)$ if and only if*

$$C|_{E_\lambda} : E_\lambda \longrightarrow \mathbb{R}^m \quad \text{is injective for every } \lambda \in \mathcal{L}. \tag{4.1}$$

Proof. Write $u_0 = \sum a_\lambda$, $u_1 = \sum b_\lambda$, with coefficients in E_λ. Then

$$y(t) = \sum_{\lambda \in \mathcal{L}} \left[\cos(\sqrt{\lambda}\, t) C a_\lambda + \frac{\sin(\sqrt{\lambda}\, t)}{\sqrt{\lambda}} C b_\lambda \right].$$

A finite collection of sine and cosine functions at distinct positive frequencies is linearly independent on any nonempty interval. To see this, express it in exponentials with distinct exponents $\pm i\sqrt{\lambda}$; differentiating a vanishing combination through the required finite number of orders at an interior point yields a Vandermonde system with nonzero determinant. Hence $y = 0$ implies $Ca_\lambda = Cb_\lambda = 0$ for every λ. Condition (4.1) then gives the zero state. Conversely, a nonzero $v \in E_\lambda \cap \ker C$ supplies the invisible solution $v \cos(\sqrt{\lambda} t)$. □

Thus two time signals can distinguish many distinct frequencies. The obstruction is the kernel within a single eigenspace, not the total number of modes. A simple eigenmode needs only $C\phi \neq 0$; an eigenspace of dimension greater than m cannot be resolved by m channels.

Choose an orthonormal eigenbasis for V_N, let D have columns $C\phi_j$, and let $\Lambda = \operatorname{diag}(\lambda_j)$. For $z = (q, q')$, define

$$M = \begin{pmatrix} 0 & I \\ -\Lambda & 0 \end{pmatrix}, \qquad B_0 = \begin{pmatrix} D & 0 \end{pmatrix}.$$

The equations are $z' = Mz$, $y = B_0 z$.

Corollary 4.2 (Reconstruction formula and conditioning)**.** *Under* (4.1)*, the matrix*

$$G_T = \int_0^T e^{tM^*} B_0^* B_0 e^{tM}\, dt$$

is positive definite, and

$$z(0) = G_T^{-1} \int_0^T e^{tM^*} B_0^* y(t)\, dt. \tag{4.2}$$

If y is replaced by $y + \eta$, the least-squares reconstruction obeys

$$\|z_{\text{rec}} - z(0)\|_2 \le \lambda_{\min}(G_T)^{-1/2} \|\eta\|_{L^2(0,T)}.$$

Proof. Let $Oz = B_0 e^{tM} z$. Then $G_T = O^*O$, so $\langle G_T z, z\rangle = \|Oz\|_{L^2}^2$. Theorem 4.1 is exactly the positivity assertion. Applying O^* to the data proves (4.2). The operator $G_T^{-1}O^*$ satisfies

$$(G_T^{-1}O^*)(G_T^{-1}O^*)^* = G_T^{-1},$$

which gives its stated norm and the error estimate. □

The displayed estimate uses Euclidean coefficient norm. For comparison across cutoffs, use energy coordinates $(\Lambda^{1/2} q, q')$ and form the Gramian in those coordinates. Positivity for every fixed cutoff does not imply a positive lower bound independent of the cutoff. Unmodeled higher modes enter a truncated reconstruction as an additional data residual.

5 A known-profile inverse source theorem

We now assume zero initial state and

$$u_{tt} + Au = b\, a(t), \qquad y(t) = Cu(t), \tag{5.1}$$

with known b and unknown a. To give simple sufficient regularity hypotheses, assume $b \in D(A)$ and $C \in \mathcal{L}(V, \mathbb{R}^m)$. Define

$$k(t) = CS(t)b.$$

By (2.1), $y = k * a$, where convolution here is causal:

$$(k * a)(t) = \int_0^t k(t-s)a(s)\,ds.$$

Lemma 5.1 (Kernel regularity)**.** *Under these assumptions, $k \in C^2([0,T];\mathbb{R}^m)$,*

$$k(0) = 0, \qquad k'(0) = Cb, \qquad k''(t) = -CA^{1/2}\sin(tA^{1/2})b.$$

For $a \in L^2(0,T)$, the data belong to $H^2(0,T;\mathbb{R}^m)$ and satisfy

$$y(0) = y'(0) = 0, \qquad y'' = Cb\,a + k'' * a. \tag{5.2}$$

Proof. The spectral theorem gives $S'(t)b = P(t)b$ and $S''(t)b = -A^{1/2}\sin(tA^{1/2})b$. These derivatives are continuous in V: applying $A^{1/2}$ to the second derivative gives $-A\sin(tA^{1/2})b$, whose norm is at most $\|Ab\|_H$. Spectral dominated convergence justifies differentiation and continuity, since $b \in D(A)$. Apply C. For a smooth input, differentiation of the Volterra integral proves (5.2). Young's inequality for each of k, k', k'' and density of smooth inputs extend the formula and the H^2 bound to L^2 inputs. □

Theorem 5.2 (Source reconstruction with sharp Sobolev loss)**.** *Assume $Cb \neq 0$. Set $\ell v = (Cb)\cdot v/|Cb|^2$ and $q(t) = \ell k''(t)$. For every $a \in L^2(0,T)$, the data uniquely determine a through*

$$a + q * a = \ell y''. \tag{5.3}$$

If $M_q = \max_{[0,T]} |q|$, then

$$\|a\|_{L^2(0,T)} \le e^{M_q T}\|\ell\|\,\|y''\|_{L^2(0,T)}. \tag{5.4}$$

The scalar projected map $a \mapsto \ell y$ is a bounded isomorphism from $L^2(0,T)$ onto

$$\{z \in H^2(0,T) : z(0) = z'(0) = 0\}.$$

For every $0 \le s < 2$, there is no constant K_s such that

$$\|a\|_{L^2(0,T)} \le K_s\|y\|_{H^s(0,T;\mathbb{R}^m)} \quad \text{for all } a \in L^2(0,T). \tag{5.5}$$

Proof. Apply ℓ to (5.2). Let Q be convolution with q on $(0,T)$. For $n \ge 1$, the n-fold convolution satisfies

$$|q^{*n}(t)| \le M_q^n \frac{t^{n-1}}{(n-1)!}, \qquad \|Q^n\|_{L^2\to L^2} \le \frac{(M_qT)^n}{n!}.$$

The first estimate follows by induction, and the second by Young's inequality. Hence $\sum_{n\ge 0}(-Q)^n$ converges in operator norm, is $(I+Q)^{-1}$, and has norm at most e^{M_qT}. This proves uniqueness and (5.4).

Given a scalar $z \in H^2$ with the two zero traces, solve $(I+Q)a = z''$. The resulting projected data and z have the same second derivative and initial traces, so they coincide. This proves surjectivity; boundedness of the forward map follows from Lemma 5.1.

Finally, the forward map is bounded from L^2 to H^2. Its composition with the compact embedding $H^2(0,T) \hookrightarrow H^s(0,T)$ is compact for $s < 2$. A lower bound such as (5.5) would give a bounded inverse on its range and make the identity on the infinite-dimensional input space compact, a contradiction. □

Example 5.3 (A complete one-mode calculation)**.** Let $A\phi = \lambda\phi$ and choose $b = \phi$ with $d = C\phi \neq 0$. The solution has the form $u = \phi v$, where

$$v'' + \lambda v = a, \quad v(0) = v'(0) = 0, \quad y = dv.$$

Therefore

$$v = \frac{d \cdot y}{|d|^2}, \qquad a = \frac{d \cdot (y'' + \lambda y)}{|d|^2}.$$

For example, on $(0,\pi)^2$, take $\phi = \sin x_1 \sin x_2$, so $\lambda = 2c^2$. Average $\partial_{x_1}\phi$ over the square of half-width h centred at $(\pi/4, \pi/2)$, where $0 < h < \pi/4$. Direct integration gives

$$C\phi = \frac{1}{\sqrt{2}} \left(\frac{\sin h}{h} \right)^2 \neq 0.$$

The theorem thus applies to a bounded patch sensor, not just an ideal point sensor. Although the square has corners, its positive Dirichlet realization satisfies all abstract hypotheses used in this section.

Recovering the input also recovers the full vibration. In fact Proposition 2.1 gives, for two inputs and their zero-initial-state solutions,

$$\sup_{0 \le t \le T} \|(u^{(1)} - u^{(2)}, u_t^{(1)} - u_t^{(2)})(t)\|_{\mathcal{H}} \le \sqrt{T}\, \|b\|_H \|a_1 - a_2\|_{L^2(0,T)}.$$

Combining this with (5.4) yields a stable reconstruction of the entire state in the energy norm from the H^2 observation norm.

Theorem 5.2 is a stable inverse in an H^2 data topology. Numerical differentiation of raw noisy L^2 data is not stable. A direct regularized formulation is

$$\min_{a \in L^2(0,T)} \{\|\mathcal{K}a - y^\delta\|_{L^2}^2 + \alpha \|a\|_{L^2}^2\}, \qquad \alpha > 0,$$

where $\mathcal{K}a = k * a$. Its unique minimizer solves $(\mathcal{K}^*\mathcal{K} + \alpha I)a = \mathcal{K}^* y^\delta$. This is an implementation option; no noise-dependent convergence rate is claimed here.

6 Uniform pressure on a circular membrane

Let $\Omega = B_R(0)$, take $b = 1$, and extend an input $a \in L^2(0,S)$ by zero outside $(0,S)$. The displacement solves

$$u_{tt} - c^2 \Delta u = a(t), \quad u|_{\partial B_R} = 0, \quad u(0) = u_t(0) = 0. \tag{6.1}$$

Rotational invariance and uniqueness imply that u is radial. Let $r = |x_*| > 0$, $e_* = x_*/r$, and $0 < \varepsilon < \min(r, R - r)$. We observe

$$y_{r,\varepsilon}(t) = C_{r,\varepsilon} u(t), \qquad C_{r,\varepsilon} v = \frac{1}{\pi\varepsilon^2} \int_{B_\varepsilon(x_*)} e_* \cdot \nabla v(x)\, dx. \tag{6.2}$$

The full vector average points along e_* by reflection symmetry. Thus its two coordinates carry only one independent signal in this model.

The hypotheses of Theorem 5.2 fail: $1 \notin D(A)$, and the gradient of the smooth constant function is zero. One cannot apply formal powers of $-\Delta$ to 1 while ignoring the Dirichlet operator domain.

Lemma 6.1 (Disk mean identity)**.** *Suppose v is smooth near $\overline{B_\varepsilon(x_*)}$ and $\Delta v = \mu^2 v$. Then*

$$\frac{1}{\pi\varepsilon^2}\int_{B_\varepsilon(x_*)} v(x)\,dx = \frac{2I_1(\mu\varepsilon)}{\mu\varepsilon}\,v(x_*), \tag{6.3}$$

with the factor interpreted as 1 at $\mu = 0$.

Proof. Let $m(\rho)$ be the circular average of v at radius ρ about x_*. Averaging the polar-coordinate equation removes the angular derivative and gives

$$m'' + \rho^{-1}m' = \mu^2 m, \qquad m(0) = v(x_*), \quad m'(0) = 0.$$

The regular solution is $m(\rho) = I_0(\mu\rho)v(x_*)$. Integrate $2\rho m(\rho)/\varepsilon^2$ from 0 to ε, using $(zI_1(z))' = zI_0(z)$. Both the regular solution and this derivative identity follow immediately from the power series

$$I_0(z) = \sum_{j\geq 0}\frac{(z^2/4)^j}{(j!)^2}, \qquad I_1(z) = \sum_{j\geq 0}\frac{(z/2)^{2j+1}}{j!(j+1)!}.$$

This proves (6.3), including its limit at zero. $\square$

For a causal function g, write $\widehat{g}(s) = \int_0^\infty e^{-st}g(t)\,dt$ for $s > 0$.

Proposition 6.2 (Exact transfer function)**.** *For* (6.1)–(6.2),

$$\widehat{y}_{r,\varepsilon}(s) = H_{r,\varepsilon}(s)\widehat{a}(s), \quad H_{r,\varepsilon}(s) = -\frac{1}{cs}\frac{I_1(sr/c)}{I_0(sR/c)}\frac{2I_1(s\varepsilon/c)}{s\varepsilon/c}. \tag{6.4}$$

Proof. The Laplace-transformed equation is $(s^2 - c^2\Delta)\widehat{u} = \widehat{a}(s)$. Its radial solution, regular at the origin and zero at radius R, is

$$\widehat{u}(\rho, s) = \frac{\widehat{a}(s)}{s^2}\left(1 - \frac{I_0(s\rho/c)}{I_0(sR/c)}\right). \tag{6.5}$$

The denominator is positive for real $s > 0$ by the series for I_0. The solution can also be verified by direct substitution; uniqueness follows from positivity of $s^2 + A$. Its radial derivative at r is

$$-\frac{\widehat{a}(s)}{cs}\frac{I_1(sr/c)}{I_0(sR/c)}.$$

Every Cartesian derivative of $\widehat{u}$ solves $\Delta v = (s/c)^2 v$ in the interior, since the forcing is spatially constant. Apply Lemma 6.1 to $e_* \cdot \nabla\widehat{u}$. This yields (6.4). The Laplace transformations are justified by the energy bound for a compactly supported L^2 input and the boundedness of the patch observation. $\square$

The impulse response is the bounded continuous function

$$k_{r,\varepsilon}(t) = C_{r,\varepsilon}S(t)1, \quad t \geq 0,$$

extended by zero for $t < 0$. Boundedness follows from $\|S(t)1\|_V \leq \|1\|_H$. In particular $y = k_{r,\varepsilon} * a$.

Proposition 6.3 (Positive-real asymptotic)**.** *For fixed r, ε as above, set*

$$\delta = \frac{R - r - \varepsilon}{c}, \qquad K_{r,\varepsilon} = \sqrt{\frac{2cR}{\pi r\varepsilon^3}}.$$

As real $s \to +\infty$,

$$H_{r,\varepsilon}(s) = -K_{r,\varepsilon}e^{-\delta s}s^{-5/2}(1 + O(s^{-1})). \tag{6.6}$$

Proof. Use the standard asymptotic $I_\nu(z) = e^z(2\pi z)^{-1/2}(1+O(z^{-1}))$ for fixed ν and positive real $z \to \infty$ [4]. The first Bessel ratio in (6.4) contributes $\sqrt{R/r}\, e^{-s(R-r)/c}(1+O(s^{-1}))$. The last factor contributes $\sqrt{2/\pi}\,(s\varepsilon/c)^{-3/2}e^{s\varepsilon/c}(1+O(s^{-1}))$. Multiplication gives (6.6). □

At fixed $s>0$, the aperture factor tends to 1 as $\varepsilon \downarrow 0$, giving the ideal point transfer

$$H_{r,0}(s) = -\frac{1}{cs}\frac{I_1(sr/c)}{I_0(sR/c)}.$$

Its positive-real leading power after removal of the delay is s^{-1}, not $s^{-5/2}$. These two limiting statements are not uniform in both s and ε. At $r=0$, symmetry gives exactly zero averaged vector slope; formulas that divide by r cannot be used at the centre.

7 An exact finite-time uniqueness statement for uniform pressure

For a nonzero causal function g, let $\alpha(g) = \inf \operatorname{supp} g$, where support is understood in the essential, or distributional, sense. Put $\alpha(0) = +\infty$.

Lemma 7.1 (The first support time)**.** *The finite-aperture impulse response satisfies*

$$\alpha(k_{r,\varepsilon}) = \delta = \frac{R-r-\varepsilon}{c}.$$

Proof. The function $v(t) = S(t)1$ solves the homogeneous wave equation with initial displacement zero and initial velocity 1. At any point whose backward cone has not reached the boundary, $v(x,t) = t$. Here is the local energy argument. Put $w = v - t$, choose $\overline{B_\rho(x)} \subset \Omega$, and consider

$$E_\rho(\tau) = \frac{1}{2}\int_{B_{\rho-c\tau}(x)} (|w_t|^2 + c^2|\nabla w|^2)\,dz.$$

For smooth solutions the wave equation and the moving-boundary differentiation formula give

$$E_\rho'(\tau) = \int_{\partial B_{\rho-c\tau}(x)} \left(c^2 w_t \partial_\nu w - \frac{c}{2}(|w_t|^2 + c^2|\nabla w|^2)\right) d\sigma \le 0.$$

Since $E_\rho(0) = 0$, the derivatives vanish throughout the shrinking ball; the zero initial displacement then gives $w=0$ in the cone. Smooth local approximation gives the same conclusion for energy solutions. This does not require the initial velocity to have zero boundary trace. Therefore $\nabla v = 0$ on the aperture whenever $0<t<\delta$, so $k=0$ there.

Suppose k also vanished on $(0, \delta+\eta)$ for some $\eta>0$. Since k is bounded, its Laplace transform would satisfy

$$|H_{r,\varepsilon}(s)| \le \|k\|_\infty \frac{e^{-(\delta+\eta)s}}{s}.$$

This contradicts the nonzero asymptotic (6.6). Hence no larger initial zero interval exists, proving the result. □

External theorem (Titchmarsh). If $f, g \in L^1(\mathbb{R})$ are nonzero and their supports are bounded below, then

$$\alpha(f*g) = \alpha(f) + \alpha(g). \tag{7.1}$$

We use this classical support theorem as an external analytic input. A proof is given in [3]. Notice that it prohibits cancellation from increasing the first support time of a convolution; the elementary inclusion of supports alone would not suffice.

Theorem 7.2 (Sharp observation time for uniqueness)**.** *Fix $S > 0$ and let inputs range over $L^2(0, S)$, extended by zero elsewhere. The map*

$$a \longmapsto y_{r,\varepsilon}|_{(0,T)}$$

is injective if and only if

$$T \geq S + \delta. \tag{7.2}$$

This is an exact-data uniqueness statement, not a stability estimate.

Proof. Consider a difference input a producing zero data on $(0, T)$. To apply (7.1), set $k_1(t) = e^{-t}k(t)$ and $a_1(t) = e^{-t}a(t)$ for $t \geq 0$, zero otherwise. Both are in $L^1(\mathbb{R})$, and $k_1 * a_1 = e^{-t}(k * a)$. Exponential weighting does not change supports. If $a \neq 0$, Theorem (7.1) and Lemma 7.1 give

$$T \leq \alpha(k * a) = \delta + \alpha(a).$$

When $T \geq S + \delta$, this implies $\alpha(a) \geq S$. An $L^2(0, S)$ function supported at or after S is zero, a contradiction. This proves injectivity, including equality in (7.2).

If $T < S + \delta$, choose a nonzero smooth input supported in $(\max(0, T - \delta), S)$. The support inclusion for causal convolution and $\alpha(k) = \delta$ imply that its output vanishes on $(0, T)$. Thus injectivity fails. □

More locally, zero data on $(0, T)$ force $a = 0$ almost everywhere on $(0, T - \delta)$ when $T > \delta$. This makes the delay obstruction explicit. No finite-time conclusion in this section is inferred merely from all-time Laplace-transform injectivity.

Even when (7.2) holds, there is no uniform L^2-to-L^2 inverse estimate. Indeed, the integral kernel $k(t - s)$, with its causal zero extension, is bounded on $(0, T) \times (0, S)$. The observation map is therefore Hilbert–Schmidt and hence compact. As before, a compact operator on an infinite-dimensional input space cannot be bounded below. The disk example thus gives both a precise uniqueness theorem and a precise instability statement.

8 Smooth spatial averaging and loss of Sobolev stability

Return to a smooth bounded domain and the uniform source $b = 1$. Let $w \in C_c^\infty(\Omega)$, fix a unit vector e, and set

$$Cu = \int_\Omega w\, e \cdot \nabla u \, dx, \qquad \psi = -e \cdot \nabla w.$$

Then $Cu = \langle \psi, u \rangle_H$ for $u \in V$; we use real Hilbert spaces in this proof.

Theorem 8.1 (Infinite smoothing for a compact smooth weight)**.** *The causal impulse response $k(t) = CS(t)1$, extended by zero for $t < 0$, belongs to $C^\infty(\mathbb{R})$ and satisfies $k^{(n)}(0) = 0$ for every $n \geq 0$. For any finite $S, T > 0$ and any finite $q \geq 0$, the observation map*

$$\mathcal{K} : L^2(0, S) \longrightarrow H^q(0, T), \qquad a \longmapsto (k * a)|_{(0,T)}$$

is compact. There is consequently no estimate

$$\|a\|_{L^2(0,S)} \leq K \|\mathcal{K}a\|_{H^q(0,T)}$$

valid for all inputs with a finite constant K.

Proof. Since ψ is compactly supported and smooth, $\psi \in D(A^m)$ for every m. Self-adjointness gives $k(t) = \langle S(t)\psi, 1\rangle_H$, which is smooth in t by spectral differentiation on ψ. Its even derivatives vanish at zero because they contain a sine factor. Its odd derivatives are

$$k^{(2m+1)}(0) = (-1)^m \langle A^m \psi, 1\rangle_H.$$

For $m = 0$, this is the integral of a compactly supported directional derivative and is zero. For $m \geq 1$, $A^m\psi = (-c^2\Delta)^m\psi$ is compactly supported and has zero integral by integration by parts. All initial derivatives vanish, so extension by zero is smooth across $t = 0$.

Extend a by zero and write $y(t) = \int_0^S k(t-s)a(s)\,ds$. On the compact difference interval $[-S, T]$, all derivatives of k are bounded. Differentiation under this fixed integral gives a bounded map into $H^n(0,T)$ for every integer n. Choose $n > q$ and use the compact embedding $H^n \hookrightarrow H^q$ to prove compactness. A bounded inverse on the range would contradict compactness of the identity on $L^2(0, S)$. □

This theorem neither asserts nor denies injectivity for every weight. For example, a rotationally symmetric weight centred on a uniformly driven disk gives identically zero vector slope. Other weights may yield an injective convolution map after a sufficient observation time, while still lacking every finite-order Sobolev stability estimate above. The effect is tied to both the uniform source and the compact smooth weight. A Gaussian with nonzero tails up to the membrane boundary is not a compactly supported interior weight and is not covered by the theorem.

9 Scope and further questions

The main results form a complete basic account of three different questions:

1. **Which initial states are visible?** Theorem 4.1 gives the finite-mode criterion, and Corollary 4.2 gives a reconstruction and noise estimate. Proposition 3.3 explains why uniform infinite-dimensional energy stability cannot follow for bounded finite-channel sensors.
2. **Can a known-profile input be reconstructed stably?** Theorem 5.2 gives an explicit Volterra inverse and a sharp two-derivative Sobolev threshold under $b \in D(A)$ and $Cb \neq 0$.
3. **What happens for uniform pressure?** Propositions 6.2–6.3 and Theorem 7.2 give an exact finite-aperture transfer and a sharp uniqueness time. Theorem 8.1 exhibits a separate obstruction to finite-order Sobolev stability for smooth compact weights.

The 5/2 issue. The factor $s^{-5/2}$ in (6.6) suggests a first-arrival kernel proportional to $(t-\delta)_+^{3/2}$. A sharp fractional-order reconstruction theorem would require a time-domain expansion with a controlled remainder and the appropriate causal trace spaces. Positive-real Laplace asymptotics alone do not establish such a theorem. No 5/2-derivative stability assertion is made here, and no arbitrary-long-time frequency-domain lower bound is inferred from (6.6).

Corners and other domains. The elementary operator arguments extend once a positive self-adjoint wave operator and a bounded observation have been specified. On a bounded Lipschitz domain, the closed form $c^2 \int \nabla u \cdot \nabla v$ with form domain H_0^1 defines the canonical Dirichlet realization. One must use its actual operator domain, which need not be $H^2 \cap H_0^1$ on a nonconvex polygon. A conic metric or an interface introduces additional domain or transmission choices. Such geometry is not needed for the present results; no diffraction or nonsmooth-domain observability theorem is claimed.

Optical nonlinearity. The inverse theorems concern the calibrated linear observation C. Formula (1.2) records the underlying nonlinear map in one ideal geometry. A theorem for the exact nonlinear optical data would need its own hypotheses and estimates; linear injectivity alone is not a substitute. Likewise, recovering electrical loudspeaker input would require an acoustic transfer model beyond the pressure forcing used here.

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